

Class 12: Genome informatics

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Section 4: Population Scale Analysis [HOMEWORK]

One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale. So, you processed about ~230 samples and did the normalization on a genome level. Now, you want to find whether there is any association of the 4 asthma-associated SNPs (rs8067378...) on ORMDL3 expression.

This is the final file you got (https://bioboot.github.io/bggn213_W19/class-material/rs8067378_ENSG0000017). The first column is sample name, the second column is genotype and the third column are the expression values.

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes. Hint: The `read.table()`, `summary()` and `boxplot()` functions will likely be useful here. There is an example R script online to be used ONLY if you are struggling in vein. Note that you can find the median value from saving the output of the `boxplot()` function to an R object and examining this object. There is also the `median()` and `summary()` function that you can use to check your understanding.

The dataset included 462 total samples, with genotype counts of 108 A/A, 233 A/G, and 121 G/G. The median expression levels were 31.25 for A/A, 25.06 for A/G, and 20.07 for G/G. These results indicate that ORMDL3 expression is highest in individuals with the A/A genotype and lowest in those with the G/G genotype.

```
bob <- read.table("BunchOfGenes")
head(bob)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169

```
5 NA18870 G/G 18.25141
6 NA11993 A/A 32.89721
```

```
nrow(bob)
```

```
[1] 462
```

```
table(bob$geno)
```

```
A/A A/G G/G
108 233 121
```

```
tapply(bob$exp, bob$geno, median)
```

```
      A/A      A/G      G/G
31.24847 25.06486 20.07363
```

```
summary(bob)
```

sample	geno	exp
Length:462	Length:462	Min. : 6.675
Class :character	Class :character	1st Qu.:20.004
Mode :character	Mode :character	Median :25.116
		Mean :25.640
		3rd Qu.:30.779
		Max. :51.518

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3? Hint: An example boxplot is provided overleaf – yours does not need to be as polished as this one.

The boxplot demonstrates that individuals with the A/A genotype exhibit the highest levels of ORMDL3 expression, whereas those with the G/G genotype show the lowest levels. This pattern suggests that the rs8067378 SNP is linked to variation in ORMDL3 expression, indicating that this SNP likely influences gene expression.

```
library(ggplot2)
```

```
ggplot(bob) +  
  aes(geno, exp, fill = geno) +  
  geom_boxplot(notch = TRUE)
```

